Willian Vieira PhD Candidate

< + > Quantitative ecologist | I love to automate stuff

EDUCATION

Current | 2017

PhD. Candidate, Ecology
Integrative Ecology Lab - Université de Sherbrooke, Canada

- Developing demographic models to better understand how trees will face climate change
- Integral Projection Models | mathematical modeling | Hierachical models | Bayesian statistics | Quantitative thecniques

2016 | 2015

Masters 2, Agroecology and Resource Management
École Nationale Supérieure des Sciences Agronomiques de Bordeaux (BSA), Bordeaux, France

- Thesis: “Modelling the dispersion of weed species in agricultural landscapes”
- Process-based modelling | R | Sensitivity analysis

2015 | 2010

BSc in Agronomy
Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONSULTING EXPERIENCE

2022 | 2020

Biostatistician
Environment and Climate Change Canada (ECCC)

- Boreal Optimal Sampling Strategy for birds monitoring in Quebec
- Scripting spatial analysis | spatially stratified sampling | R
- Code: , Report:

RESEARCH EXPERIENCE

2018

Research visiting
National Research Institute for Agriculture, Food and Environment INRAe - EcoSystèmes et Sociétés En Montagne — Grenoble, France

- Collaboration on the mathematical formulation of how forest management affect forest dynamics
- Analytical analysis | Modelling | ShinyApp | R package

2017

Mitacs Internship
Ouranos — Montréal, CA

- Forest dynamic models | Climate change models

2016

Master 2 internship
French National Institute for Agricultural Research (INRA - UMR 1347 Agroecologie) — Dijon, France

- Modelling weed dispersion in agricultural landscapes to improve the sustainable management of weeds

CONTACT

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Publications

QUANTITATIVE SKILLS

Bayesian statistics, Hierarchical models, Random Forest, Mathematical modelling, Regression

PROGRAMMING SKILLS

Data analysis: R, Python, Stan, HPC, QGIS

Data base: SQLite

Automation/reports: markdown, pandoc, Quarto, LateX, bash, makefile

Web: HTML/CSS, ShinyApp

DevOps (CI/CD): Git, GitHub, Docker, GitHub Actions

LANGUAGES

Portuguese Native
French Full Professional
English Full Professional

*Made with **pagedown**.*

The source code is available at github.com/willvieira/cv.

Last updated on 2024-12-01.

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2015 | 2014

Undergraduate Researcher
Group of Research in Tropical forests — Florianópolis, Brazil
(NPFT - Universidade Fedaral de Santa Catarina)
- Characterize the dynamics of tropical forest
- Data analysis | Statistics | R | data management | field work

2014 | 2012

Undergraduate Researcher
Laboratory of Plant Developmental Physiology and Genetics — Florianópolis, Brazil
(LFDGV - Universidade Fedaral de Santa Catarina)
- Quantify the genetic diversity of tropical trees
- Lab work | PCR | microsatellites markers | genotyping

TEACHING EXPERIENCE

2021

Biodiversity Modelling Summer School
Bios^2 summer school — Sherbrooke, CA
- Teaching assistant and lectured for graduate students
- Evaluating ecological models using Bayesian statistics

2020 | 2019

Methods in computational ecology
BIO500 - Départament de biologie, Université de Sherbrooke — Sherbrooke, CA
- Teaching assistant and lectured for undergraduate students
- Tools for reproducible science | SQL | Git | R | LaTeX | Markdown | Makefile

2020

Software and Data Carpentry workshop
Bios^2 training for graduate students — Online
- The Unix Shell
- Data Management with SQL for Ecologists

2019 | 2018

QCBS R Workshops
Université de Sherbrooke & Université du Québec à Rimouski — Sherbrooke & Rimouski, CA
- Instructor to graduate students
- Workshops: Introduction to R | Data management | Data Visualization | Control flow | Linear models

2019

Probability and statistics
GCI145 - Faculté de génie, Université de Sherbrooke — Sherbrooke, CA
- Teaching assistant and lectured for undergraduate students
- Probability theory | Distribution | Descriptive statistics | Hypothesis testing | Linear models

2018 | 2017

Soil ecology
Départament de biologie - Université de Sherbrooke — Sherbrooke, CA
- Teaching assistant and lectured for undergraduate students
- Lab | Covered methods to determine soil texture

2018

Modelisation de la dynamique forestier du Quebec face au changement climatique
Départament de biologie, Université de Sherbrooke — Sherbrooke, CA
- Guest lecturer in Plant Ecology (Presentation) to undergraduate students
- Covered modelling approaches in ecology | model construction and applications

2018 **Introduction to scientific programming**
BIO109 - Départament de biologie, Université de Sherbrooke — Sherbrooke, CA
- Teaching assistant for undergraduate students
- Pogramming concepts | Data management | Control flow | R language

TRAINING

2022 **Data visualization and biodiversity modelling**
Summer school - Bios² — Magog, CA
- Business intelligence | data user profiles | Conceptualize interactive dashboard | Figma

2020 **Mathematical Modelling in Ecology and Evolution**
Workshop - Bios² — Remote
- Developing equations | Finding equilibria | analyzing stability | Maxima

2019 **Stage-based demographic models in ecology, evolution and conservation biology**
NERC Advanced Training - University of Oxford — Oxford, UK
- Theory and practice | Matrix population models | Integral Projection Models

2019 **Biodiversity Modelling**
Summer school - Université de Sherbrooke — Magog, CA
- Theory and practice on different biodiversity modelling techniques
- Statistics | Differential equations | Stochastic simulations | Climate change models | uncertainty tracking

2017 **Data-driven ecological synthesis**
DDES summer school - Université de Montréal — Montréal, CA
- Good practices in scientific computing
- Data management | Mathematical tools | Data analysis tools | Parallel computing | Rreproducible science

2017 **Bayesian Statistics for Ecologists**
Summer school - Université de Sherbrooke — Magog, CA
- Theory and practice using Stan | Probability theory | Likelihood | MCMC | Hierarchical modelling | Model comparison

PUBLICATIONS

2024 **Paying colonization credit with forest management could accelerate the range shift of temperate trees under climate change**
Ecological Modelling doi
- Vieira, W., Boulangeat, I., Brice, M., Bradley, R.L., Gravel, D.

2020 **Moderate disturbances accelerate forest transition dynamics under climate change in the temperate–boreal ecotone of eastern North America**
Global Change Biology doi
- Brice, M.H., Vissault, S., Vieira, W., Gravel, D., Legendre, P., Fortin, M.J.

2020 **Genetic and demographic aspects of Varronia curassavica Jacq. in a heterogeneous coastal ecosystem**
Anais da Academia Brasileira de Ciências
• Hoeltgebaum, M.P., Lauterjung, M.B., Montagna, T., Candido-ribeiro, R., Vieira, W., Bernardi, A.P., Cristofolini, C., Reis, M.S.D.

SELECTED TALKS AND POSTERS

2020 **Lab: {STManaged} R package**
Biodiversity Modelling summer school — Magog, CA
• Demonstration of my R package to be used during the intensive course (Presentation)

2019 **Using Integral Projection Models to assess how Contemporary Niche Theory predicts trees range limits**
British Ecological Society Annual Meeting — Belfast, UK
• (Poster)

2018 **The effect of forest management on forest distribution under climate change**
QCBS annual symposium — Montréal, CA
• (Presentation)

2017 **Can forest management help increase the rate of forest migration northward to follow climate change?**
European Geosciences Union General Assembly — Vienna, AT
• (Poster)